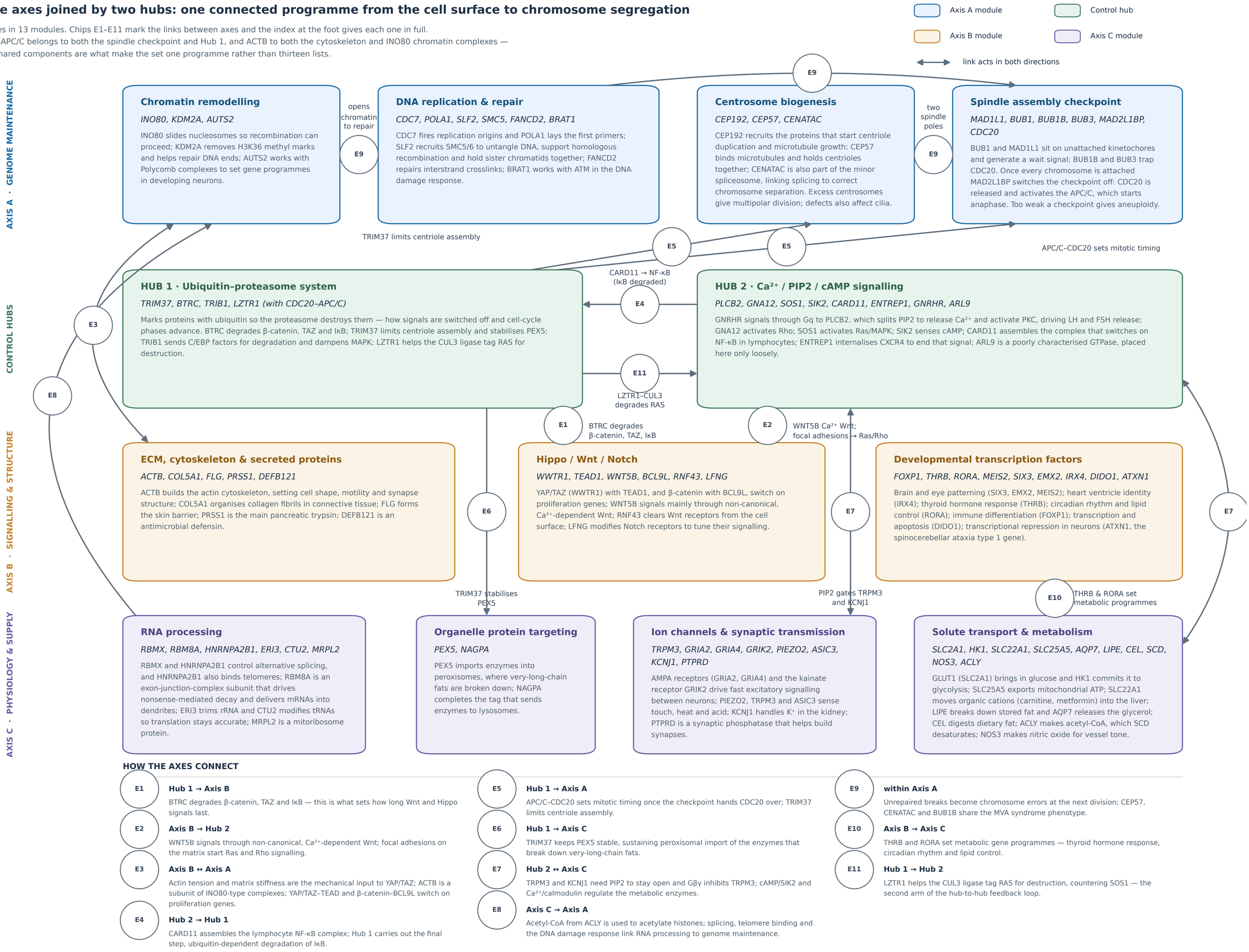
Three axes joined by two hubs: one connected programme from the cell surface to chromosome segregation

76 genes in 13 modules. Chips E1–E11 mark the links between axes and the index at the foot gives each one in full.
CDC20–APC/C belongs to both the spindle checkpoint and Hub 1, and ACTB to both the cytoskeleton and INO80 chromatin complexes — these shared components are what make the set one programme rather than thirteen lists.



HOW THE AXES CONNECT

E1 — Hub 1 → Axis B
BTRC degrades β-catenin, TAZ and IκB — this is what sets how long Wnt and Hippo signals last.

E2 — Axis B → Hub 2
WNT5B signals through non-canonical, Ca²⁺-dependent Wnt; focal adhesions on the matrix start Ras and Rho signalling.

E3 — Axis B ↔ Axis A
Actin tension and matrix stiffness are the mechanical input to YAP/TAZ; ACTB is a subunit of INO80-type complexes; YAP/TAZ–TEAD and β-catenin–BCL9L switch on proliferation genes.

E4 — Hub 2 → Hub 1
CARD11 assembles the lymphocyte NF-κB complex; Hub 1 carries out the final step, ubiquitin-dependent degradation of IκB.

E5 — Hub 1 → Axis A
APC/C–CDC20 sets mitotic timing once the checkpoint hands CDC20 over; TRIM37 limits centriole assembly.

E6 — Hub 1 → Axis C
TRIM37 keeps PEX5 stable, sustaining peroxisomal import of the enzymes that break down very-long-chain fats.

E7 — Hub 2 ↔ Axis C
TRPM3 and KCNJ1 need PIP2 to stay open and Gβγ inhibits TRPM3; cAMP/SIK2 and Ca²⁺/calmodulin regulate the metabolic enzymes.

E8 — Axis C → Axis A
Acetyl-CoA from ACLY is used to acetylate histones; splicing, telomere binding and the DNA damage response link RNA processing to genome maintenance.

E9 — within Axis A
Unrepaired breaks become chromosome errors at the next division; CEP57, CENATAC and BUB1B share the MVA syndrome phenotype.

E10 — Axis B → Axis C
THRB and RORA set metabolic gene programmes — thyroid hormone response, circadian rhythm and lipid control.

E11 — Hub 1 → Hub 2
LZTR1 helps the CUL3 ligase tag RAS for destruction, countering SOS1 — the second arm of the hub-to-hub feedback loop.